Sec: OSR.IIT_*COSC(MODEL-A&APEX) | GTA-1(P1) | Date: 22-03-26
Time: 3 Hrs | 2020_P1 | Max. Marks: 198

KEY SHEET
MATHS

1	**A**	2	**C**	3	**B**	4	**D**	5	**A**
6	**A**	7	**BCD**	8	**CD**	9	**ACD**	10	**ABD**
11	**AC**	12	**AC**	13	**9**	14	**3**	15	**192**
16	**11**	17	**0**	18	**0**				

PHYSICS

19	**D**	20	**B**	21	**A**	22	**D**	23	**A**
24	**B**	25	**ABCD**	26	**ABD**	27	**ABCD**	28	**AB**
29	**AB**	30	**AB**	31	**1.43**	32	**2.14**	33	**14**
34	**0.78 TO 0.82**	35	**4.50**	36	**87**				

CHEMISTRY

37	**D**	38	**A**	39	**C**	40	**A**	41	**D**
42	**B**	43	**AB**	44	**ABD**	45	**ACD**	46	**ABC**
47	**ABCD**	48	**ABC**	49	**1.60**	50	**200**	51	**50**
52	**69.30**	53	**0.17 TO 0.22**	54	**6**				

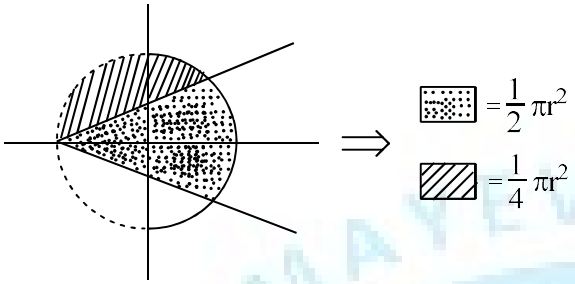
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SOLUTIONS

MATHS

1. No. of derangements of the word 'SUDHEER' such that E_1, E_2 should not occupy either E_1 or E_2 place

$= D_7 - [D_6 + D_6 + D_5]$

2. 

$= \frac{1}{2}\pi r^2$

$= \frac{1}{4}\pi r^2$

3. $2x^3 + 3x^2 - 1 = t \Rightarrow \int \frac{dt}{t^{3/2}(t-1)^{1/2}}$

4. $\int_0^x f(x).t^2 dt = x.\int_0^x t.f(t).dt$

$\frac{f(x).x^2}{3} = \int_0^x t.f(t).dt$

$\frac{2x}{3}.y + \frac{x^2}{3}.\frac{dy}{dx} = x.y$

$x^2\frac{dy}{dx} = xy$

$\frac{dy}{y} = \frac{dx}{x}$

$\ell n|y| = \ell n|x| + c$

$y = k.x$

5. $A^{10} = I$, A and B are commutative, $B^2 = -I$

$\sum_{r=1}^{r}\left(B^2 - BA^{9r} - A^rB + I\right) = -\sum_{r=1}^{4}\left(A^T + A^{9r}\right).B$

$\Rightarrow -\left(A + A^2 + ... + A^9 - A^5\right).B = 0$

$I + A + A^2 + ... + A^9 = 0$ (or)

6. let 'd' denoted dotted, 'D' denotes digits each cube has 4 choices : Rd, RD, Bd, BD

Assume no of dice as

Rd = x, RD = y

Bd = z, BD = w

If A & B are independent $\frac{x}{\cancel{3}} = \left(\frac{x+y}{\cancel{3}}\right)\left(\frac{x+z}{3}\right) \Rightarrow 3x = (x+y)(x+z)$

$0 \le x + y + z \le 3,\ x, y, z, w \in \{0,1,2,3\}, x + y + z + w = 3$

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x	y	z
0	0	1,2,3
0	1,2,3	0
0	0	0
1	2	0
1	0	2
2	1	0
2	0	1
3	0	0

7. For C, D options consider

$$f(x)=\sin\left(\frac{\pi}{2}\left[x+\frac{1}{2}\right]\right)$$

$$f(-x)=\sin\left(\frac{\pi}{2}\left[-x+\frac{1}{2}\right]\right)=\sin\left(\frac{\pi}{2}\left[1-\left(x+\frac{1}{2}\right)\right]\right)=\sin\left(\frac{\pi}{2}-\frac{\pi}{2}\left[-\left(x+\frac{1}{2}\right)\right]\right)$$

$$\text{If } x+\frac{1}{2}\notin z \Rightarrow f(-x)=\sin\left(\frac{\pi}{2}-\frac{\pi}{2}\left(-1-\left[x+\frac{1}{2}\right]\right)\right)$$

$$=\sin\left(\pi+\frac{\pi}{2}\left[x+\frac{1}{2}\right]\right)=-\sin\left(\frac{\pi}{2}\left[x+\frac{1}{2}\right]\right)$$

$$\text{If } x+\frac{1}{2}\in z;\ x+\frac{1}{2}=n\ n\in z \Rightarrow f(-x)=\sin\left(\frac{\pi}{2}-\frac{\pi}{2}[-n]\right)=\sin\left(\frac{\pi}{2}-\frac{\pi}{2}(-[n])\right)$$

$$=\sin\left(\frac{\pi}{2}(1+[n])\right)$$

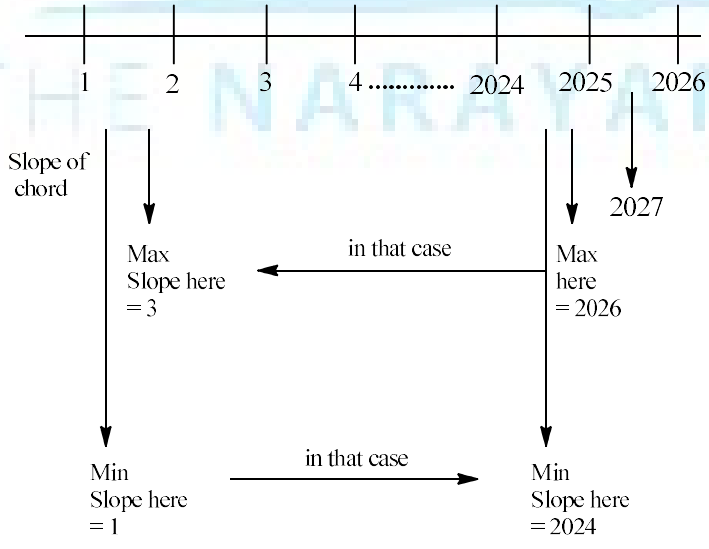
$$f(x)+f(-x)=\sin\left(\frac{\pi}{2}[n]\right)+\cos\left(\frac{\pi}{2}[n]\right)$$

$$n=x+\frac{1}{2}=4k+\frac{1}{2}+\frac{1}{2}=4k \text{ or } 4k+1 \Rightarrow f(x)+f(-x)=1$$

8. Since $f''>0\Rightarrow f'$ is increasing $\frac{f(x+1)-f(x)}{(x+1)-x}$ is +ve integer. $\forall x\in N$ between $[x,x+1]$ at some point $f'=+ve$ integer. $\forall x\in N$

Since f' is increasing, slope of chord should also Increase

$$\frac{f(2026)-f(2025)}{1}=2027$$



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9. $S_1 = 0$

$\frac{\sqrt{2}x}{\cos\theta} - \frac{y}{\sin\theta} = 1$ and compare

10. $\frac{f(x)+1+1}{3} \geq \sqrt[3]{f(x)}$

$h(x) = \int_0^x f(t)dt - x$

$\int_0^1 \left(\frac{3f}{1+2f} - 1\right)dx \leq 0$

$\int_0^1 \left(\frac{f-1}{1+2f}\right)dx \leq 0$

11. $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, $x = 0$, $y + 3z = 0$ or $3y - z = 0$

12. A = {1,2, 3, …..,n}

B = $\{1^2, 2^2, 3^2, ..., n^2\}$

C = $\{1^3, 2^3, 3^3, ..., n^3\}$

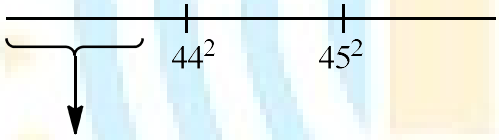
Frequency of the unique mode will be always 3 and mode should be 1

next common element = $2^{2\times3}$

$\therefore S = \{1, 2, 3, ..., 63\}$

For median we need to find 2025th, 2026th observations when arranged in increasing order

Lets try to arrange then on a line



1, 2, 3, …..,1936

$1^2, 2^2, ..., 43^2$

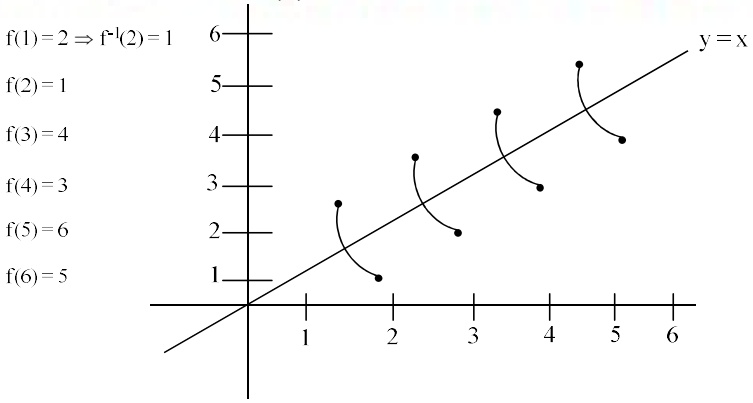
44^2 will be 1980th number

2026 – 1980 = 46

$\therefore$ 2025th observation = 1936 + 46 = 1982

$\therefore$ median = $\frac{1982+1983}{2} = 1982.5$

13. x = 1, 2, 3, 4, 5 and f(x) = x for 4 values



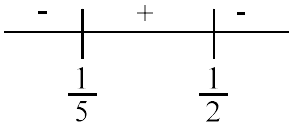
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14. as $n \to \infty$ $12x\dfrac{\left(n^2+7x-1\right)}{n^2+10x^2} \to 12x$

$n^2+7x-1 > n^2+10x^2$

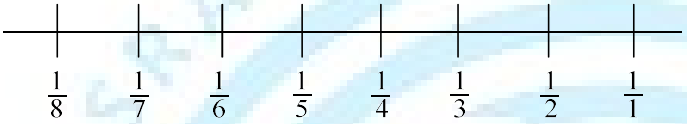
$-10x^2+7x-1>0 \Rightarrow -10x^2+5x+2x-1$

$(1-5x)(2x-1)>0$



For $x > \dfrac{1}{2}$ or $x < \dfrac{1}{5}$; if tends to $(12x)^-$

For $\dfrac{1}{5} < x < \dfrac{1}{2}$; if tends to $(12x)^+$



$\left[\dfrac{12}{m}\right] = \dfrac{12}{m} \Rightarrow \dfrac{12}{m}$ is integer

The values of m should be such that $\dfrac{12}{m}$ is integer and $\left(\dfrac{n^2+7x-1}{n^2+10x^2}\right) \to 1^+$

15. $\begin{vmatrix} x_1+iy_1 & x_1-iy_1 & 1 \\ x_2+iy_2 & x_2-iy_2 & 1 \\ x_3+iy_3 & x_3-iy_3 & 1 \end{vmatrix} = \begin{vmatrix} x_1+iy_1 & -2iy_1 & 1 \\ x_2+iy_2 & -2iy_2 & 1 \\ x_3+iy_3 & -2iy_3 & 1 \end{vmatrix}$

$C_2 \to C_2 - C_1$ $\qquad$ $C_1 \to C_1 - C_2$

$= -2i\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$ = 4 area of triangle = $4.\dfrac{abc}{4R}$

abc = 48R = 48 x 4

$\left|(z_1-z_2)(z_1-z_3)(z_3-z_1)\right| =$ abc

Or

Method-II:

Replace $\overline{z_2} = \dfrac{4}{z_2}$ and factorize

16. $A_1 = \dfrac{a+b}{2}$ $A_2 = \dfrac{a+\frac{a+b}{2}}{2} = \dfrac{3a+b}{4}$ $A_3 = \dfrac{a+\frac{3a+b}{4}}{2} = \dfrac{7a+b}{8} \Rightarrow A_n = \dfrac{\left(2^n-1\right)a+b}{2^n}$

$H_1 = \dfrac{2ab}{a+b}$ $H_2 = \dfrac{2a.\left(\frac{a+b}{2}\right)}{a+\frac{a+b}{2}} = \dfrac{2a(a+b)}{3a+b}$

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$$H_3 = \frac{2a.\left(\frac{3a+b}{4}\right)}{a+\frac{3a+b}{4}} = \frac{2a(3a+b)}{7a+b}$$

$$H_n = \frac{2a.\left(\left(2^{n-1}-1\right)a+b\right)}{\left(2^n-1\right)a+b}$$

$$\text{If } A_{n+1} = H_n \quad \frac{\left(2^{n+1}-1\right)a+b}{2^{n+1}} = \frac{2a\left(\left(2^{n-1}-1\right)a+b\right)}{\left(2^n-1\right)a+b}$$

$$\left(2^{n+1}-1\right)\left(2^n-1\right)a^2+\left(2^{n+1}+2^n-2\right)ab+b^2 = 2.2^{n+1}.\left(2^{n-1}-1\right)a^2+2^{n+1}.2ab$$

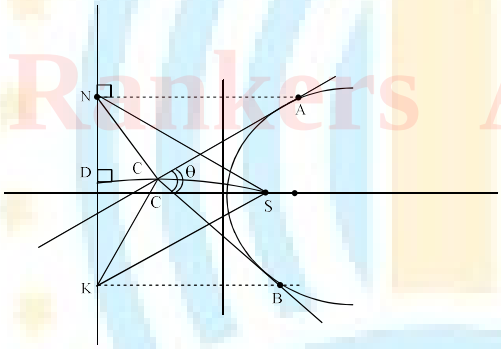
$$b^2+\left(2^{n+1}+2^n-2^{n+2}-2\right)ab+\left(2^{n+2}-2^n-2^{n+1}+1\right)a^2=0$$

$$2^{n+2}-2^n-2^{n+1} = 2^n.\left[2^2-1-2\right] = 2^n$$

$$b^2-\left(2^n+2\right).ab+\left(2^n+1\right)a^2=0$$

$$\frac{b}{a} = 1 \text{ or } 2^n+1$$

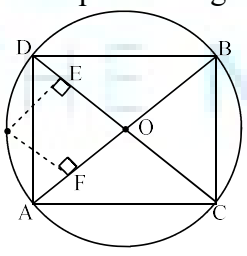
17. AS = AN $\Rightarrow$ tangent at A is perpendicular bisector of NS
$\Rightarrow$ Circumcentre is C



$$\frac{CD}{CK} = \frac{CD}{CS} = \cos 60° \Rightarrow \frac{CS}{CD} = 2$$

Locus is hyperbola with eccentricity 2

18. Lines pass through centre of the circle



$$\frac{1}{2}PA.PB = \frac{1}{2}PF.AB$$
$$\frac{1}{2}PC.PD = \frac{1}{2}PE.CD$$
$$\Rightarrow 14^2+48^2$$
$$r = 5$$

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19. The force experienced by A is due to infinite number of incidence and reflection. The situation is shown in the diagram

$$F=\frac{I}{c}+\frac{0.5I}{c}\times 2+\frac{0.5^2I}{c}\times 2+\frac{0.5^2I}{c}\times 2+.....$$

$$=\frac{I}{c}+\frac{2I}{c}\left[\frac{1}{2}+\frac{1}{2^2}+\frac{1}{2^3}+....\right]$$

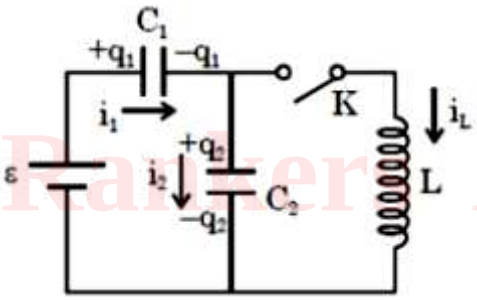
$$=\frac{I}{c}+\frac{2I}{c}\times\left[\frac{1/2}{1-1/2}\right]=\frac{3I}{c}$$

20. $\frac{W}{Q_\ell+W}=\eta_\ell \quad \frac{W}{Q_\ell}=\eta_2$

$$\frac{W}{Q_\ell}=\ell-\frac{300}{T}$$

$$\frac{1}{1+\frac{T}{T-300}}=\frac{1200-T}{1200}$$

21.



qc_2 maximum Þ $\frac{dq_2}{dt}=0$ Þ $i_2=0$ Þ $i_L=i_1$

qc_1 maximum Þ $\frac{dq_1}{dt}=0$ Þ $i_1=0$ power delivered by battery = 0

qc_2 maximum Þ V_L maximum Þ $\frac{Ldi_L}{dt}$ maximum Þ $\frac{d^2i_L}{dt^2}=0\Rightarrow i_2=0$

KVL in left loop Þ $\varepsilon-\frac{q_1}{C_1}-\frac{q_2}{C_2}=0$ Þ $\frac{i_1}{C_1}+\frac{i_2}{C_2}=0\Rightarrow\frac{i_1}{i_2}=\frac{C_1}{C_2}$

22. The peak height and width are determined by the relationships $A_{max}\approx\frac{F_0}{b\omega_0}$ and $\Delta\omega\approx\frac{b}{m}$.

1. **Peak height**: Since $b_1=2b_2$, the damping for the first oscillator is greater. Therefore, $A_1=\frac{1}{2}A_2$. The peak is lower.

2. **Peak width**: The width $\Delta\omega$ is proportional to $\frac{b}{m}$. Since $\frac{b_1}{m_1}=\frac{2b_2}{4m_2}=\frac{1}{2}\left(\frac{b_2}{m_2}\right)$, the width of the first oscillator is half that of the second. The peak is **narrower**.

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23. $\Sigma F_z = 0$

$(T + dT) + \mu g dz - T = 0$

$dT = -\mu g dz$ ….. (i)

Also $T = \mu v^2$

$dT = d\mu v^2 + 2v dv\, d\mu$

As v is independent of z

$dv = 0$

$dT = v^2 d\mu$ ….. (ii)

From equation (1) and (2) we get $\mu \int \frac{d\mu}{\mu} = -\frac{g}{v^2}\int_0^z dz$ or $\mu = \mu_0 e^{-(g/v^2)_z}$

24. $A = A_0\left(\frac{\sin\beta}{\beta}\right)$ or $\frac{A}{A_0} = \frac{\sin\beta}{\beta}$

Now $\beta = \frac{\pi a \sin\theta}{\lambda} = \frac{\pi}{2}$

$\therefore \frac{A}{A_0} = \frac{\sin\left(\frac{\pi}{2}\right)}{\left(\frac{\pi}{2}\right)} = \frac{2}{\pi}$

25. $0.8 = \frac{hC}{500} - \phi = \frac{1240}{500} - \phi$

$\phi = 1.68 eV$

$1.2 = \frac{hC}{\lambda} - 1.68$

$2.88 = \frac{1240}{\lambda}$

$\lambda = \frac{1240}{2.88}$

$f = \frac{C}{\lambda} = \frac{3\times10^8}{430.5\times10^{-9}}$

26. Since piston is movable $\Rightarrow$ force on piston in equilibrium must be zero.

$\therefore$ Pressure in both chamber must be same

By conserving number of moles

$\frac{PV}{T} = \frac{P_1V_1}{T}$ ….. (i)

$\& \frac{2P\times 2V}{T} = \frac{P_f(3V - V_f)}{T}$ ….. (ii)

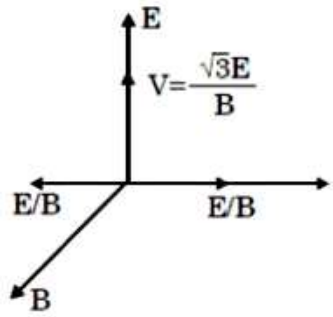
Solving (i) & (ii), $V_f = \frac{3V}{5}$ (volume of chamber – I)

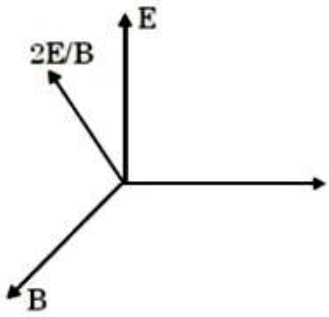
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Volume of chamber – II will be $3V - V_f = \frac{12V}{5}$

Also $P_f = \frac{5}{3}P$

27.





Now the particle will perform a super position of uniform motion and circular path

28. At $v = \frac{v_0}{2}$, the energy equation is $K_0 = \frac{1}{4}K_0 + \frac{1}{2}Li^2$.

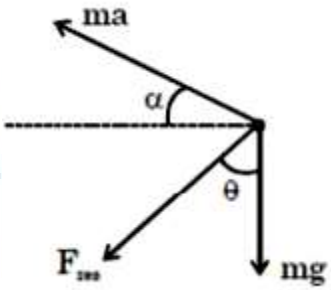
- $\frac{1}{2}Li^2 = \frac{3}{4}\left(\frac{J^2}{2m}\right) = \frac{3J^2}{8m}$
- $i^2 = \frac{3J^2}{4Lm} \Rightarrow i = \sqrt{\frac{3J^2}{4Lm}}$ (Option B)

3. Flux Linkage

From $L\frac{di}{dt} = B\ell\frac{dx}{dt}$, integrating gives $Li = B\ell d$.

- $d = \frac{Li}{B\ell} = \frac{L}{B\ell}\sqrt{\frac{3J^2}{4Lm}}$
- $d = \sqrt{\frac{3J^2L}{4B^2\ell^2m}}$ (Option A)

29. Consider an element on the surface



Angle θ which resultant force makes with vertical will be the angle made by surface with horizontal $\tan\theta = \frac{a\cos\alpha}{g - a\sin\alpha}$

(A) $a = 0$ $\tan\theta = 0$

(B) $a = \frac{g}{\sqrt{2}}$ $\tan\theta = 1$ $\theta = 45°$

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(C) $a=\frac{1}{4}\frac{g}{\sqrt{2}}$ $\tan\theta=\frac{1}{7}$ $\theta=8°$

30. $g-\omega^2R=\frac{g}{3}$

$\omega^2R=\frac{2g}{3}$

$\left(\frac{2\pi}{60\times60}\right)^2\times R=\frac{2}{3}\times\frac{GM}{R^2}$

$\frac{4\pi}{3600}\times\frac{3\times2}{2G}\frac{GM}{4/3R^3}=P$

$P=\frac{2\pi}{3600\times3600\times2\times(20/3)\times10^{-6}}=\frac{10^6\pi}{192}\approx5000\,p\,kg/m^3$

$\omega^2R=\frac{2g}{3}=\frac{2}{3}\frac{GM}{R^2}$

$\omega^2(R+h)=\frac{GM}{(R+h)^2}$

$\left(1+\frac{h}{R}\right)=\frac{3}{2}\frac{R^2}{(R+h)^2}\Rightarrow1+h=\sqrt{\frac{3}{2}}\Rightarrow h=\left(\sqrt{\frac{3}{2}}-1\right)R$

$gh=g-\omega^2R\cos^2\theta=g-\frac{2g}{3}\times\frac{1}{4}=\frac{5g}{6}$

31. On the ring of mass m the forces acting are its weight and normal reactions due to the two hoops symmetrically.

Hence $mg-2N\sin\alpha=ma$ ------ (1)

Assuming ring moves along y-axis the position of ring is (0, $(0,R(1+\sin\alpha))$. The x coordinate of right hoop is $R\cos\alpha$

Applying contact constraint the horizontal acceleration of CM of a hoop and acceleration of ring are related as

$a_h\cos\alpha=a\sin\alpha$

$\Rightarrow a_h=a\tan\alpha$ _____ (2)

Angular acceleration of hoop can be found out by $=\frac{NR\cos\alpha}{2MR^2}=\frac{N\cos\alpha}{2MR}$

$\therefore a_h=\frac{N\cos\alpha}{2M}$ _____ (3)

By solving questions (1), (2) & (3) we get

$a=\frac{mg}{m+4M\tan^2\alpha}$

Substituting values we get $a=\frac{10}{7}m/s^2$

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32. From folding symmetry & applying series parallel combination we get that the equivalent resistance of circuit between the leads as $\frac{7R}{15}$

Hence current is $I=\frac{15U}{7R}$

33. If acceleration of CM of pulley is a_p the constraint relation is

$a=3a_p$ _____ (1)

T is tension in string connecting A

T^1 is total tension in strings connecting B to roof.

Then applying newton's second law we get

$mg-T=ma$ _____ (2)

$Mg+T-T^1=Ma_p$ _____ (3)

$MgR+T(3R)=(I+MR^2)\alpha$ ____ (4)

$a_p=R\alpha$ ____ (5)

Solving the 5 equations we get

$$a=\frac{3g(M+3m)}{M+9m+\frac{I}{R^2}}$$

Substituting values we get p + q + r +s = 14.

34. $T=2\pi\sqrt{\frac{\ell}{|\vec{g}-\vec{w}|}}=0.8s$, where $|\vec{g}-\vec{w}|=\sqrt{g^2+w^2-2gw\cos\beta}$

35. The power radiated by an accelerating electron is given by the Larmor formula:

$$\frac{dE}{dt}=-\frac{2e^2w^2}{3c^3}$$

For harmonic oscillations $x=A\cos(\omega t)$, the average acceleration squared is

$(\omega)^2=\frac{1}{2}\omega^4A^2$.

Since $E=\frac{1}{2}m\omega^2A^2$, the energy decay follows:

$$\frac{dE}{dt}=-\gamma E,\ \text{where}\ \gamma=\frac{2e^2\omega^2}{3mc^3}$$

$$E(t)=E_0e^{-\gamma t}\Rightarrow t=\frac{\ln(E_0/E)}{\gamma}=\frac{3mc^3\ln(\eta)}{2e^2\omega^2}$$

36. $\frac{3F}{4M}$ $\frac{7F}{8}$

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CHEMISTRY

37. $NH_4NO_2 \xrightarrow[60-70^\circ C]{\Delta} N_2 + 2H_2O$

$NH_4NO_2 \xrightarrow[200-250^\circ C]{\Delta} N_2O + 2H_2O$

39. For Ideal Gas PV = nRT

$\therefore n \propto P$ at constant T & V

$\because \text{mole} = \dfrac{\text{Mass}}{\text{Molar mass}}$

$\text{For gas X}: \dfrac{10}{M_x} \propto 2\,\text{atm}$ ..... (1)

For gas X & Y : $\dfrac{10}{M_x} + \dfrac{80}{M_Y} \propto 6\,\text{atm}.....(2)$

From (2) – (1)

$\dfrac{80}{M_Y} \propto 4$ ..... (3)

On dividing (1) by (3) $\dfrac{M_Y}{8M_X} = \dfrac{1}{2}$

$\therefore \dfrac{M_Y}{M_X} = 4$ .... (4)

$\because V_{mp} = \sqrt{\dfrac{2RT}{M}} \Rightarrow V_{mp} \propto \dfrac{1}{\sqrt{M}}$

$\therefore \dfrac{(V_{mp})_X}{(V_{mp})_Y} = \sqrt{\dfrac{M_Y}{M_X}} \Rightarrow \sqrt{\dfrac{4}{1}} = \dfrac{2}{1}$

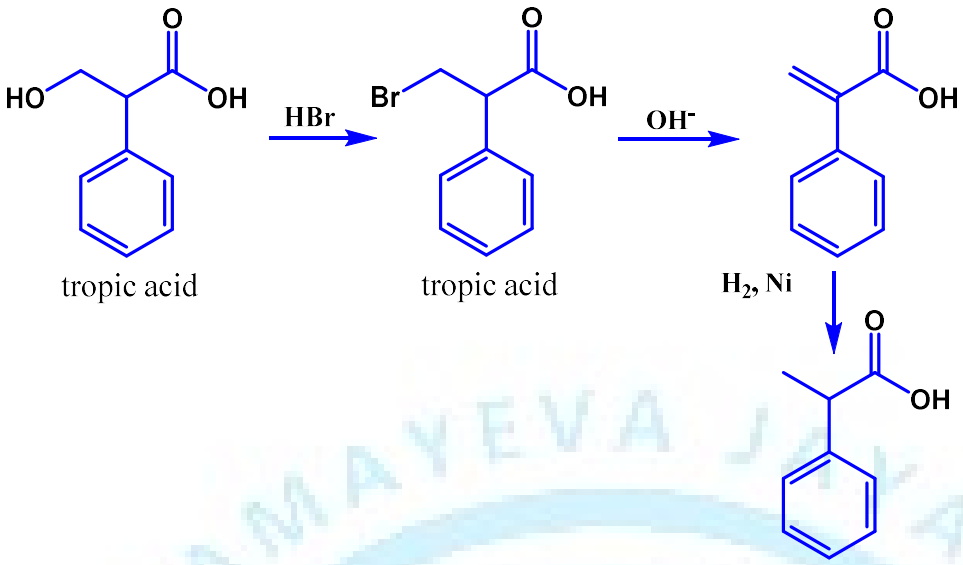
40. $K_4[Fe(CN)_6] + H_2O_2 + H_2SO_4 \rightarrow K_3[Fe(CN)_6] + K_2SO_4 + H_2O$

$K_3[Fe(CN)_6] + H_2O_2 + OH^- \rightarrow K_4[Fe(CN)_6]Z(CrCl_3) \xrightarrow{H_2O_2OH^\ominus}$ Yellow solution $\left(CrO_4^{2-}\right)$

41. Alkali solution dissolve impure alumina, CO_2 gas is added to complex to change pH & regenerate alumina

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43.



44. $T_m = 300k, \Delta H_{fus} = 6KJ/mol$

a) $\Delta S_{fus} = \frac{\Delta H_{fus}}{T} = \frac{6000}{300} = 20\,J\,mol^{-1}K^{-1}$

b) Heating solid $250K \rightarrow 300K$ $\Delta S = \int_{250}^{300} \frac{C_P(solid)}{T} dT$

c) At equilibrium solid and liquid co-exist

d) At eq. $\Delta G = \Delta H - T\Delta S$

$\Delta G = 0,\ \Delta H = T\Delta S$

45. $\underset{s}{AgCl(s)} \rightleftharpoons \underset{(s-x)}{Ag^+} + \underset{s}{Cl^-}$ $K_{sp} = 10^{-10}M^2$

$\underset{s-x}{Ag^+} + \underset{\substack{c=2\\c-2x}}{2NH_3(aq)} \rightleftharpoons \underset{x}{\left[Ag(NH_3)_2\right]^+}(aq)$ $K_f = 10^8$

$10^{-10} = (s-x)s$ .... (1)

$10^8 = s$

$(s-x)(c-2s)^2$ .... (2)

$10^{-2} = \frac{s^2}{(c-2s)^2}$

$s = 0.1$

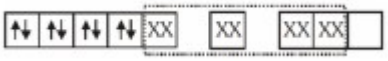
$2 - 2s$

$s = 0.2 - 0.25$

$1.25 = 0.2$

$s = \frac{0.2}{1.2} = \frac{1}{6}$

46. A) $Pt(78)[Xe]_{54}\,4f^{14}6s^1 5d^9 Pt^{2+} 6s^0 5d^8$

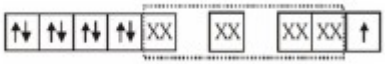


It has dsp^2 hybridization, square planar shape.

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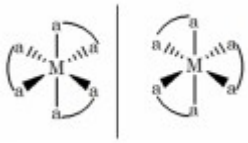
$Cu(29):[Ar]4s^1 3d^{10}$

$Cu^{2+}:[Ar]4s^0 3d^9$

$[Cu(NH_3)_4]^{2+}$ 

It has dsp^2 hybridisation, square planar shape.

B) $\left[Fe(C_2O_4)_3\right]^{3-}$ is $M(a-a)_3$ type complex



The mirror images are non superimposable. Hence, optically active.

C) Synergic bonding in metal carboynyl $\uparrow$, CO bond strength decreases.

D) $\left[NiCl_2(PPh_3)_2\right]$ is tetrahedral due to steric reasons, so NO geometrical isomers

47. For 1st order $t_{99\%} = \frac{1}{K_{eff}}\ell n\left(\frac{100}{1}\right) - \frac{2\ln 10}{K_{eff}}$

$$t_{99\%} = \frac{1}{K_{eff}}\ell n\left(\frac{100}{10}\right) - \frac{\ell n 10}{K_{eff}}$$

$$\frac{t_{99\%}}{t_{99\%}} = 2$$

$$t_{90\%} = 10 = \frac{\ell n 10}{K_{eff}} \Rightarrow K_{eff} = \frac{\ell n 10}{10}\sec^{-1}$$

$$t_{1/2} = \frac{\ell n(2)}{K_{eff}} = \frac{10\ell n(2)}{\ell n(10)} = 10 \times 0.3010 = 3.01\sec$$

$$\frac{t_{15/16}}{t_{3/4}} = \frac{\ell n(16)}{\ell n(4)} = 2$$

49. $\frac{x}{m} = K \times C^{1/n}$

$$\log\left(\frac{x}{m}\right) = \log K + \frac{1}{n}\log C$$

$$\log 4 = \log K + \frac{1}{n}\log 10$$

$$0.6 = \log K + \frac{1}{n} \ldots (1)$$

$$\log 10 = \log K + \frac{1}{n}\log 16$$

$$1 = \log K + \frac{1}{n} \times 1.2 \ldots (2)$$

Equation (2) – equation (1)

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$$0.4 = \frac{1}{n} \times (0.2) \Rightarrow n = 0.5 \text{ and } \log K = -1.4$$

$$\log \frac{x}{m} = \log K + \frac{1}{n} \times \log C = -1.4 + 2 \times \log 32 = -1.4 + 3 = 1.6$$

$$\frac{x}{m} = 10^{16}$$

$$\left(\log 2 = 0.3,\ 4\log 2 = 1.2,\ 16 = 10^{+1.2}\right)$$

50. Vessel – 1 : $(\Delta T_b)_1 = i_1 \times K_b \times \frac{w_2 / GMM_X}{w_1 / 1000}$

Vessel – 2 : $(\Delta T_b)_2 = i_2 \times K_b \times \frac{w_2 / GM\ M_Y}{w_1 / 1000}$

$$\frac{(\Delta T_b)_1}{(\Delta T_b)_2} = \frac{i_1}{i_2} \times \frac{GMM_Y}{GMM_X} = \frac{1.2}{0.6} = 2$$

$$\frac{(\Delta T_b)_1}{(\Delta T_b)_2} \times 100 = \frac{2}{1} \times 100 = 200\%$$

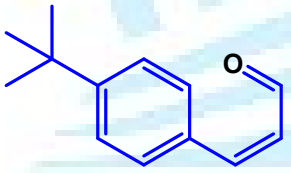
51. X = 13:- reaction 3, 4, 6 are correct and include acidic – impurities
Y = 3 :- reactions 1, 2 are correct and include basic impurities
Z = 5 :- reaction 5 is incorrect

52. $K_{eff} = K\left[H^+\right] = \frac{1}{10} \ell n\left(\frac{150-50}{150-100}\right) \Rightarrow K_{eff} = \frac{\ell n(2)}{10}$

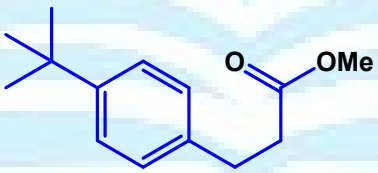
$$t_{1/2} = \frac{\ln(2)}{K_{eff}} = 10 \sec$$

$$K\left(10^{-3}\right) = \frac{0.693}{10} \Rightarrow K = 69.3 M^{-1} \sec^{-1}$$

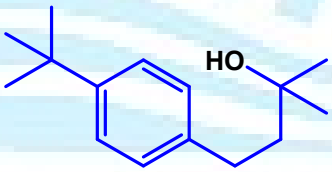
54.



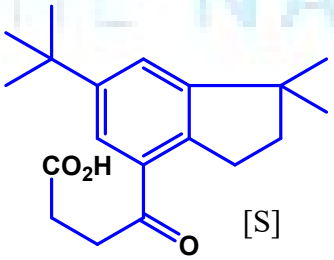
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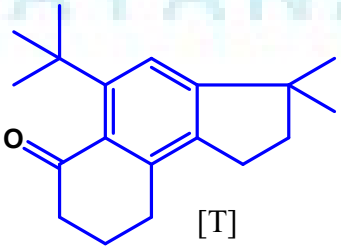
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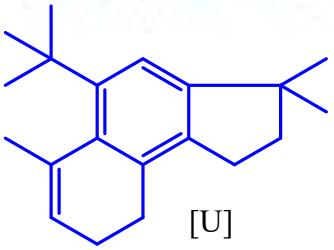
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